Mobile Application Development

Lecture 05

Todays Agenda

- Unit of Measurement
- Types of Layouts
  - View & View Groups
  - Linear Layout
  - Relative Layout
  - Constraints Layout

Units of Measurement

Sr.No	Unit & description
1	**dp** Density-independent pixel. 1 dp is equivalent to one pixel on a 160 dpi (dot per inch) screen.
2	**sp** Scale-independent pixel. This is similar to dp and is recommended for specifying font sizes
3	**pt** Point. A point is defined to be 1/72 of an inch, based on the physical screen size.
4	**px** Pixel. Corresponds to actual pixels on the screen

View

- This class represents the basic building block for user interface components.
- A View occupies a rectangular area on the screen and is responsible for drawing and event handling.
- View is the base class for *widgets*, which are used to create interactive UI components (buttons, text fields, etc.).

Using View

- All of the views in a window are arranged in a single tree.
- You can add views either from code or by specifying a tree of views in one or more XML layout files.
- There are many specialized subclasses of views that act as controls or are capable of displaying text, images, or other content.

ViewGroup

- A ViewGroup is a special view that can contain other views (called children)
- The view group is the base class for layouts and views containers
- This class also defines the ViewGroup.LayoutParams class which serves as the base class for layouts parameters
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- The base LayoutParams class just describes how big the view wants to be for both width and height. For each dimension, it can specify one of:
    - FILL_PARENT (renamed MATCH_PARENT in API Level 8 and higher), which means that the view wants to be as big as its parent (minus padding)
    - WRAP_CONTENT, which means that the view wants to be just big enough to enclose its content (plus padding)

Common Attributes of View & View Groups

Sr.No	View & description
1	**layout_width** Specifies the width of the View or ViewGroup
2	**layout_height** Specifies the height of the View or ViewGroup
3	**layout_marginTop** Specifies extra space on the top side of the View or ViewGroup
4	**layout_marginBottom** Specifies extra space on the bottom side of the View or ViewGroup
5	**layout_marginLeft** Specifies extra space on the left side of the View or ViewGroup
6	**layout_marginRight** Specifies extra space on the right side of the View or ViewGroup
7	**layout_gravity** Specifies how child Views are positioned
8	**layout_weight** Specifies how much of the extra space in the layout should be allocated to the View

Linear Layout

- *Android LinearLayout is a view group that aligns all children in either vertically or horizontally.*

Linear Layout Attributes

- Following are the important attributes specific to LinearLayout

Sr.No	Attribute & Description
1	**android:gravity** This specifies how an object should position its content, on both the X and Y axes. Possible values are top, bottom, left, right, center, center_vertical, center_horizontal etc.
2	**android:orientation** This specifies the direction of arrangement and you will use "horizontal" for a row, "vertical" for a column. The default is horizontal.
3	**android:weightSum** Sum up of child weight

Linear Layout Example